FOLD Progress

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New Model: XLM-R

- Compared to mBERT +13.8% average accuracy on XNLI, +12.3% average F1 score on MLQA, and +2.1% average F1 score on NER
- Correlates better with FSI (Pearson coef +.2, Spearman not much change)
- Phonemic Similarity takes a hit, but only Spearman (may be some problems with this calculation)

New Dataset: UN Parallel Corpus

- 6 way parallel corpus in UN official languages (ar, en, es, fr, ru, zh)
- human translated UN docs from before 2015
- Corpus with 15 languages is still in progress

Bible Dataset

compute_overlaps (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.333	0.058	-0.551	0.001	33
lex_sim	-0.800	0.000	-0.878	0.000	56
mut_int	0.499	0.000	0.550	0.000	52
pho_sim	-0.874	0.000	-0.641	0.000	465

compute_overlaps (Bible, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.536	0.001	-0.573	0.000	34
lex_sim	-0.800	0.000	-0.872	0.000	56
mut_int	0.498	0.000	0.463	0.001	52
pho_sim	-0.875	0.000	-0.523	0.000	496

kl_divergence_matrix (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.400	0.021	0.511	0.002	33
lex_sim	0.802	0.000	0.889	0.000	56
mut_int	-0.515	0.000	-0.524	0.000	52
pho_sim	0.875	0.000	0.669	0.000	465

kl_divergence_matrix (Bible, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.591	0.000	0.569	0.000	34
lex_sim	0.800	0.000	0.863	0.000	56
mut_int	-0.545	0.000	-0.576	0.000	52
pho_sim	0.878	0.000	0.543	0.000	496

coherence_fun (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.317	0.072	-0.516	0.002	33
lex_sim	-0.820	0.000	-0.913	0.000	56
mut_int	0.537	0.000	0.543	0.000	52
pho_sim	-0.895	0.000	-0.579	0.000	465

coherence_fun (Bible, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.506	0.002	-0.514	0.002	34
lex_sim	-0.857	0.000	-0.904	0.000	56
mut_int	0.411	0.002	0.399	0.003	52
pho_sim	-0.852	0.000	-0.394	0.000	496

XNLI Dataset

compute_overlaps (XNLI, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.151	0.606	-0.376	0.185	14
lex_sim	-0.943	0.000	-0.782	0.013	9
mut_int	-0.547	0.453	0.000	1.000	4
pho_sim	-0.954	0.000	-0.547	0.000	78

compute_overlaps (XNLI, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.036	0.903	0.044	0.881	14
lex_sim	-0.947	0.000	-0.842	0.004	9
mut_int	0.547	0.453	0.000	1.000	4
pho_sim	-0.950	0.000	-0.283	0.012	78

kl_divergence_matrix (XNLI, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.261	0.368	0.459	0.098	14
lex_sim	0.943	0.000	0.799	0.010	9
mut_int	0.539	0.461	-0.200	0.800	4
pho_sim	0.955	0.000	0.580	0.000	78

kl_divergence_matrix (XNLI, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.346	0.226	0.528	0.052	14
lex_sim	0.945	0.000	0.845	0.004	9
mut_int	0.358	0.642	-0.400	0.600	4
pho_sim	0.954	0.000	0.494	0.000	78

coherence_fun (XNLI, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.177	0.546	-0.352	0.217	14
lex_sim	-0.944	0.000	-0.976	0.000	9
mut_int	-0.547	0.453	0.316	0.684	4
pho_sim	-0.956	0.000	-0.573	0.000	78

coherence_fun (XNLI, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.140	0.634	-0.152	0.605	14
lex_sim	-0.956	0.000	-0.976	0.000	9
mut_int	0.547	0.453	0.000	1.000	4
pho_sim	-0.897	0.000	-0.353	0.002	78

compute_overlaps (UN 6 Way, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.826	0.085	-0.949	0.014	5
lex_sim	-1.000	0.000	-0.707	0.182	5
pho_sim	-0.981	0.000	-0.855	0.002	10

compute_overlaps (UN 6 Way, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.423	0.478	-0.158	0.800	5
lex_sim	-1.000	0.000	-0.707	0.182	5
pho_sim	-0.980	0.000	-0.758	0.011	10

kl_divergence_matrix (UN 6 Way, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.740	0.153	0.791	0.111	5
lex_sim	1.000	0.000	0.707	0.182	5
pho_sim	0.981	0.000	0.842	0.002	10

kl_divergence_matrix (UN 6 Way, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	0.718	0.172	0.791	0.111	5
lex_sim	1.000	0.000	0.725	0.165	5
pho_sim	0.980	0.000	0.851	0.002	10

coherence_fun (UN 6 Way, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.825	0.085	-0.949	0.014	5
lex_sim	-1.000	0.000	-1.000	0.000	5
pho_sim	-0.981	0.000	-0.919	0.000	10

coherence_fun (UN 6 Way, XLM-R)

metric	p_coef	p_pval	s_coef	s_pval	num_points
fsi	-0.721	0.170	-0.791	0.111	5
lex_sim	-1.000	0.000	-1.000	0.000	5
pho_sim	-0.974	0.000	-0.732	0.016	10

Problem: Autosimilarity

- When calculating correlation, should the diagonals be ignored? ie en-en, hr-hr
- Not a problem for FSI and mutual intelligence
- Phonemic and lexical similarity diagonals are 1, all fold metrics are 1
- similiarity between the same language is trivial

compute_overlaps (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	-0.800	0.000	-0.878	0.000	56
pho_sim	-0.874	0.000	-0.641	0.000	465

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	-0.610	0.000	-0.618	0.000	35
pho_sim	-0.584	0.000	-0.562	0.000	435

kl_divergence_matrix (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	0.802	0.000	0.889	0.000	56
pho_sim	0.875	0.000	0.669	0.000	465

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	0.604	0.000	0.648	0.000	35
pho_sim	0.702	0.000	0.596	0.000	435

coherence_fun (Bible, BERT)

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	-0.820	0.000	-0.913	0.000	56
pho_sim	-0.895	0.000	-0.579	0.000	465

metric	p_coef	p_pval	s_coef	s_pval	num_points
lex_sim	-0.670	0.000	-0.661	0.000	35
pho_sim	-0.623	0.000	-0.486	0.000	435